

Run 11 | OME

India

IV + controls

Target Opportunity-motivated entrepreneurship; binary 1 vs. 0

Sample 27,555 respondents **Coverage** 1 country; 2015–2022

Positive cases 2,167 (7.86%) **Case time** 55s

Train / val. / test 18,739 / 3,306 / 5,510 (68% / 12% / 20%, stratified)

Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.9877). Highest F1: Logistic Regression (0.7907). For Random Forest, adding controls changed ROC-AUC by +0.0027 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.9877	0.8563	96.37	74.42	81.99	0.7802	<1s
CatBoost	0.9877	0.8541	96.17	70.63	87.76	0.7827	11s
H2O GBM	0.9876	0.8512	96.26	74.30	80.14	0.7711	3s
Lasso	0.9867	0.8423	96.10	69.60	89.38	0.7826	12s
Ridge	0.9867	0.8437	96.24	70.70	89.15	0.7886	9s
Logistic Regression	0.9867	0.8441	96.26	70.60	89.84	0.7907	1s
Elastic Net	0.9863	0.8386	96.19	72.90	81.99	0.7717	11s
XGBoost	0.9862	0.8318	96.10	69.75	88.91	0.7817	<1s
Decision Tree	0.9845	0.7893	95.63	67.78	84.53	0.7523	<1s
Naive Bayes	0.9127	0.5737	93.01	54.58	66.05	0.5977	5s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	801.32	40.73
2	Established business owner	315.33	16.03
3	Business angel	76.41	3.88
4	Occupational status	75.25	3.82
5	Age	67.92	3.45
6	Knowing entrepreneurs	58.84	2.99
7	Household size	48.68	2.47
8	Survey year	46.55	2.37

Share is the percentage of total forest impurity importance across all 40 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Source: Completed Objective 1 batch, 27 September 2026; 11_obj1_india_iv_cv_OME. Scores and importance are taken directly from saved CSV outputs.